SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): Malawi UNDP Office
Solution Maker (SM): XCapit

1. Challenge Definition

1.1 Refined Challenge Statement

Malawi (18.2 MIO ppl) struggles with mismanagement, inefficiencies, and corruption in the distribution of development aid and public funds. Aid delivery processes often rely on manual record-keeping, centralized databases, and intermediaries, making it difficult to track resource allocation and usage. This lack of transparency undermines donor confidence, reduces community engagement, and prevents aid from reaching the intended beneficiaries, particularly in rural areas where oversight and logistical challenges are more pronounced. These issues hinder sustainable development across key sectors like health, education, and agriculture.

Over 25% of Malawi's national budget is dependent on foreign grants and donations. The inability to demonstrate effective, transparent use of these funds directly threatens the country's fiscal stability and its ability to deliver essential public services.

The urgency of this challenge is underscored by the social realities on the ground. A staggering 70% of Malawians live on less than $2.15 a day, with a significant portion of this population depending on social safety nets and subsidies for survival. This is especially true for the more than 81% of Malawians who reside in rural areas, where oversight is sparse and physical and digital access is limited. With roughly 72% of the population lacking internet access, traditional digital solutions are not viable. We need a solution that bridges this digital divide to ensure that aid truly reaches its final destination. Our goal is to not only enhance service delivery but to rebuild the fundamental trust between citizens, the government, and the international community.

1.2 Local Context

Access to digital payments is low, particularly in rural areas. Most social assistance transfers are still done in cash, leading to risks of loss, duplication, and fund diversion. Main beneficiaries include rural families, women, youth, and smallholder farmers. While World Bank and African Development Bank, and the government, have launched digital assistance programs, interoperability and trust remain significant challenges.

This challenge embraces the objective of building a new transparent, traceable and efficient aid distribution system in order to correctly manage the supporting funds at the same time that reaches the end beneficiaries as themselves without useless intermediate.

NGOs, Private Foundations, multilateral organizations and the Malawi Government work together in order to write the selection criteria, set the end beneficiaries and end up disbursing the funds through mobile wallets majorly.

Agricultural Aid and the Affordable Inputs Programme (AIP) in Malawi

For decades, Malawi has implemented agricultural input subsidy programs to boost productivity and food security. The largest of these is the Affordable Inputs Programme (AIP), introduced in 2020, is the latest iteration of these efforts, succeeding the Farm Inputs Subsidy Programme (FISP).

The AIP's primary goal is to ensure food security by helping smallholder farmers access essential inputs like fertilizers, certified seeds, and livestock at a subsidized cost. In the 2022/2023 season alone, the government allocated over MK109 billion (approximately $215 million) for this program, covering more than 70% of the cost for subsistence farmers. The program targeted millions of beneficiaries, distributing inputs across the country.

However, the program has faced persistent challenges. Reports and investigations from the Office of the Ombudsman highlight numerous issues, including:

- Financial Mismanagement: There have been significant financial discrepancies, with large sums of money paid to companies that did not deliver supplies.
- Operational Inefficiency: Beneficiaries have complained about delays in accessing inputs, inadequate stock at distribution points, and unreliable network services needed for the digital redemption process.
- Corruption: Complaints consistently mention demands for bribes by officials and other stakeholders to facilitate the purchase of inputs. There is also a lack of transparency in the selection of beneficiaries and instances of political interference.

Despite the significant financial investment by the government, these systemic issues continue to undermine the program's effectiveness and its ability to deliver inputs reliably to the intended beneficiaries.

Social Cash Program

The Social Cash Transfer Programme (SCTP)—locally known in Chichewa as Mtukula Pakhomo—is a non-conditional critical safety net for the most vulnerable, ultra-poor Malawians. By providing monthly cash transfers to over 1.3 million people annually (as of 2024), this programme helps ultra-poor families to meet their basic needs and build resilience, with the ultimate goal of building human capital and moving them out of poverty.

Social Cash Fund Breakdown

The Social Cash transfer Program targets ultra poor households under the Malawi National Social Protection Strategy 2023 to 28 (where income is less $1.25 to 2.5 a day), By the end of 2024, the SCTP reached approximately 381,735 households and over 1.7 million

individuals, including over 600,000 children, roughly ten percent of the population, disbursements are roughly between K14,919 per household and an extra K4,000 per school going child.

Telco Breakdown

A total of 11.77 million cellular mobile connections were active in Malawi in early 2024, with this figure equivalent to 55.5 percent of the total population. These subscribers are controlled by a duopoly where Airtel and Telecommunication Network Malawi (TNM) are the main players controlling close to 100% of the market share (44% and nearly 56, respectively). Each of these has a financial division which operates mobile wallets which are primary means of financial access.

1.3 Alignment with UNDP Country Office and National Development Priorities

The proposed solution directly aligns with the UNDP Malawi Country Office's strategic objectives of poverty reduction, institutional strengthening, and building economic resilience.

At a national level, the project strongly supports the government's National Development Framework by advancing two critical priorities: financial inclusion for unbanked populations and digital transformation of public services.

By building on ongoing efforts to digitize social payments, the solution introduces the Cardano blockchain as a robust layer for transparency, auditability, and scalability. This creates direct synergies with existing government strategies aimed at modernizing cash transfer programs, and complements donor-funded efforts focused on providing secure, low-tech access to finance for vulnerable communities and youth.

1.4 Relevance to UNDP Programmes and Resource Mapping

This pilot builds on an existing initiative that started under the UNICEF Innovation Fund, where we developed and field-tested an SMS-based, chain-agnostic disbursement infrastructure designed to reach beneficiaries in low-connectivity environments while maintaining strong traceability and operational control for implementing partners. The work began with an EVM-compatible implementation to enable account abstraction patterns and scalable smart-wallet orchestration, and later evolved to additional networks as we prioritized interoperability and portability across ecosystems.

From there, we expanded the same core infrastructure into UNICEF-related deployments under two main implementation lines: AidLink / Smart Pay and Shelter. AidLink / Smart Pay was piloted in Cusco, Peru (August 2024) with local partners, validating end-to-end flows for onboarding, beneficiary wallet creation using only phone numbers, and real-world spending/transfer operations without requiring smartphones or internet access. The initiative was later extended to Nairobi, Kenya (December 2025) as part of the AidLink implementation work, demonstrating the feasibility of replicating the model in a different operational context and confirming the robustness of the SMS interaction layer, disbursement logic, and monitoring/traceability requirements.

In parallel, Shelter is the name used for the Stellar-network implementation, built to operate as a complementary component to the Stellar Disbursement Platform, leveraging the same “bulk onboarding + bulk disbursement” architecture while adapting to Stellar’s ecosystem requirements. Shelter was deployed in production in October 2025, strengthening our ability to deliver the same outcomes under a chain-specific integration approach while keeping the product direction chain-agnostic.

Scope and scale. Across these initiatives, the scope consistently includes: (1) uploading and validating beneficiary lists, (2) massive wallet creation and secure linking to phone numbers, (3) bulk disbursement with configurable rules and auditability, and (4) a low-tech user experience via SMS flows (e.g., balance checks, confirmations, transfers/purchases) suitable for fragile settings. While early pilots were intentionally limited in size to validate real usage and operational processes, the architecture is designed for scale through automation of wallet creation, batched disbursements, and standardized reporting outputs for programme teams.

Timeline. The pilot is therefore not standalone: it is the continuation of a staged roadmap that includes (i) the initial UNICEF Innovation Fund build, (ii) the Peru pilot (August 2024), (iii) production deployment on Stellar as Shelter (October 2025), (iv) the Kenya AidLink implementation work (December 2025), and (v) the next step of deploying the same capabilities on Cardano, keeping the platform interoperable and adaptable across partners and regions.

Co-financing / in-kind contributions. The work leverages significant in-kind contributions from our team in the form of prior R&D, reusable components (SMS orchestration, wallet lifecycle management, bulk disbursement modules, monitoring/reporting), and accumulated implementation know-how from Peru/Kenya/Stellar production. In addition, we build on an existing collaboration network (including local partners in deployments) and reusable integrations/operational playbooks developed through these initiatives, reducing time-to-pilot and implementation risk for the Cardano deployment.

1.5 Key Stakeholders & Potential Partners

- **Government of Malawi**: Ministry of Finance- Exploratory, Ministry of Gender and Social Welfare - Exploratory, MInistry of Agriculture - Exploratory.

- **NGOs & Humanitarian Agencies**: NGOs: Givedirectly- In consultation, UNICEF- Exploratory, WFP - Exploratory

- **Communities**: Rural families and vulnerable groups.

- **Tech Partners**: Cardano Foundation, Xcapit.

1.6 Expected Impact

- Full traceability of agricultural aid.
- Financial inclusion for unbanked populations through blockchain or sms wallet.
- Reduced corruption and increased citizen trust.
- A replicable model for other African countries.

1.7 Target SDGs and Indicators

- **SDG 1**: No poverty (1.4 access to basic financial services).
- **SDG 5**: Gender equality (5.a equal access to economic resources).
- **SDG 16**: Strong institutions (16.6 effective, accountable, and transparent institutions).

1.8 Pilot Implementation Steps

Duration	Phase	Activity
Weeks 1-2	Phase 1: Preparation & System Mapping	- Beneficiary Selection**:** Work with the Ministry of Gender or AIP or partner organization coordinators to select 100 participants in a high-need area with stable cellular coverage in Dowa and Mchinji. - Cardano "Shadow Wallet" Setup - Stablecoin Selection
Weeks 3-4	Phase 2: Onboarding & Digital Literacy	- Community Sensitization - SIM-Card/Identity Linking - SMS Interaction Training**:** Conduct "Mock Transfers."
Weeks 5-6	Phase 3: Disbursement & Redemption	- Bulk Disbursement**:** Trigger the smart contract to distribute the funds from the sponsor wallet to the beneficiary wallets - Real-Time Monitoring: Provide the UNDP Country Office with a dashboard showing 100 successful transactions. - The "Last Mile" Redemption: Since beneficiaries need cash or goods, partner with a local AIP Vendor or

		Mobile Money Agent or Goods supplier.
Weeks 7-8	Phase 4: Evaluation & Audit	- Audit Trail Generation**:** Export all transaction hashes. - Beneficiary Feedback**:** Conduct a survey on the 100 participants

2. User Mapping & Testing Strategy

2.1 Primary User Persona

- **Profile**: Rural mother of three, no bank account, owns a basic feature phone.
- **Environment**: Lives in a village with limited internet, depends on farming to feed her family, and relies on local merchants for supplies.
- **Goal**: Receive social aid in a way that is safe, transparent, and usable in local shops.
- **Challenge**: Cannot access banking infrastructure or mobile money platforms that require internet or smartphones.

2.2 User Story

“As a mother in a rural farming community, I want to receive financial or agricultural aid directly in an SMS-accessible wallet on my basic phone, so that I can use it safely and transparently with local providers—without needing internet access.”

2.3 Testing Strategy

User testing will be conducted in two stages: pre-pilot validation and in-pilot / post-pilot evaluation.

Pre-pilot (before onboarding/disbursement): once a potential pilot partner confirms a target group of beneficiaries, we run a short baseline survey and a lightweight remote or in-person walkthrough of the SMS flows (receiving messages, confirming identity/phone, checking balance, and understanding basic actions). This step is used to validate language, comprehension, accessibility constraints (literacy, device type, connectivity), and to adjust wording and flow sequencing before real funds are involved.

In-pilot and post-pilot (during and after real usage): user testing continues during the pilot through (1) structured observation of first-time use (first disbursement and first transaction), (2) support logs and interaction analytics from the SMS/transaction system (errors, drop-offs, misunderstood prompts, repeated steps), and (3) a post-pilot survey and short interviews to capture satisfaction, perceived trust, ease of use, and barriers. Testing happens at key milestones: immediately after onboarding, after the first successful transaction, and at the

end of the pilot to consolidate feedback and define iteration priorities for the next deployment.

2.4 Feedback Loop Design

Feedback will be gathered through a structured loop that combines beneficiary insights, implementation partner input, and system-level evidence, and then translates them into prioritized product changes.

Gathering feedback

Beneficiaries: baseline and post-pilot surveys, short guided interviews, and (when possible) observation of first-time use. We also analyze SMS interaction patterns (drop-offs, repeated steps, misunderstood prompts) and transaction outcomes to detect friction that users may not report directly.

Stakeholders (NGOs, field teams, merchants, and programme managers): structured check-ins before and during the pilot, plus a post-pilot retrospective focused on operational workflows (beneficiary list preparation, onboarding, disbursement approvals, reporting needs, offramp/settlement constraints, and compliance/KYC realities).

Support channels: issues reported through local partners or helpdesk logs are captured with context (device type, language, network constraints, user segment) to distinguish isolated incidents from systematic issues.

Integrating feedback

All feedback is consolidated into a single backlog and tagged by: user type (beneficiary/field team/merchant/admin), flow step (onboarding, balance, transfer, purchase, settlement), severity, frequency, and risk (fund loss, user lockout, etc).

We run regular triage sessions with the pilot partner to agree priorities, define acceptance criteria, and align changes with programme timelines.

Improvements are delivered in short iteration cycles: rapid updates to SMS copy and flow logic, configuration changes, and scheduled releases for more complex upgrades. Each iteration is re-validated with a small user subset before being rolled out broadly in the pilot.

Closing the loop

After each release or operational change, we measure impact using both qualitative results (user comprehension and satisfaction) and quantitative signals (completion rates, error rates, support requests, time-to-first-transaction), and we document learnings in a pilot report to inform the next deployment.

3. Solution Overview

3.1 Solution Summary

Web3 AgriVoucher solution—extending Xcapit's SMS-enabled offline wallet

Bringing blockchain transparency and accountability to agricultural aid in partnership with Cardano Foundation, UNDP Malawi Office, and a chosen NGO.

The goal of our collaborative MVP is to deliver transparent agricultural aid and drive efficiencies that extend the reach of donor contributions. The MVP will showcase our blockchain powered partner portal & SMS-enabled offline wallet in a real-world pilot with an NGO and beneficiaries. It demonstrates account setup via SMS, voucher and fund transfers, sponsored accounts, and simple account management. A NGO partner portal highlights automated voucher distribution through smart contracts, dashboards for traceability, and claimable balances. The pilot will also test frictionless merchant transactions, confirming voucher redemptions via SMS without apps, credit, or internet access.

3.2 Core Features and Functionalities

The Voucher System manages community-distributed funds through a combination of **on-chain** (smart contracts on Cardano written in **Aiken**) and **off-chain** components— the Low-Tech Wallet (LTW) Service with SMS integration.

- Each voucher is a **UTxO locked with ADA**, controlled by a **Validator** that enforces spending conditions defined in the **Datum**.
- Community funds are first aggregated in a custodial UTxO (managed by the LTW service, no contract logic required).
- The LTW Service subdivides this UTxO into multiple voucher UTxOs, each governed by smart contract logic.
- Voucher rules:
    - Only the assigned recipient may spend their voucher during the validity window.
    - If unspent upon expiry, the steward can reclaim the funds.
    - Partial spends are supported by splitting UTxOs into merchant payment + remaining balance.
- Transactions are facilitated via **OTP-based authentication over SMS**, ensuring accessibility for users without internet.

Capabiliti	**Stakehol**	**Advantage**	**Core Functionality**	**Cardano Technical Use Case**

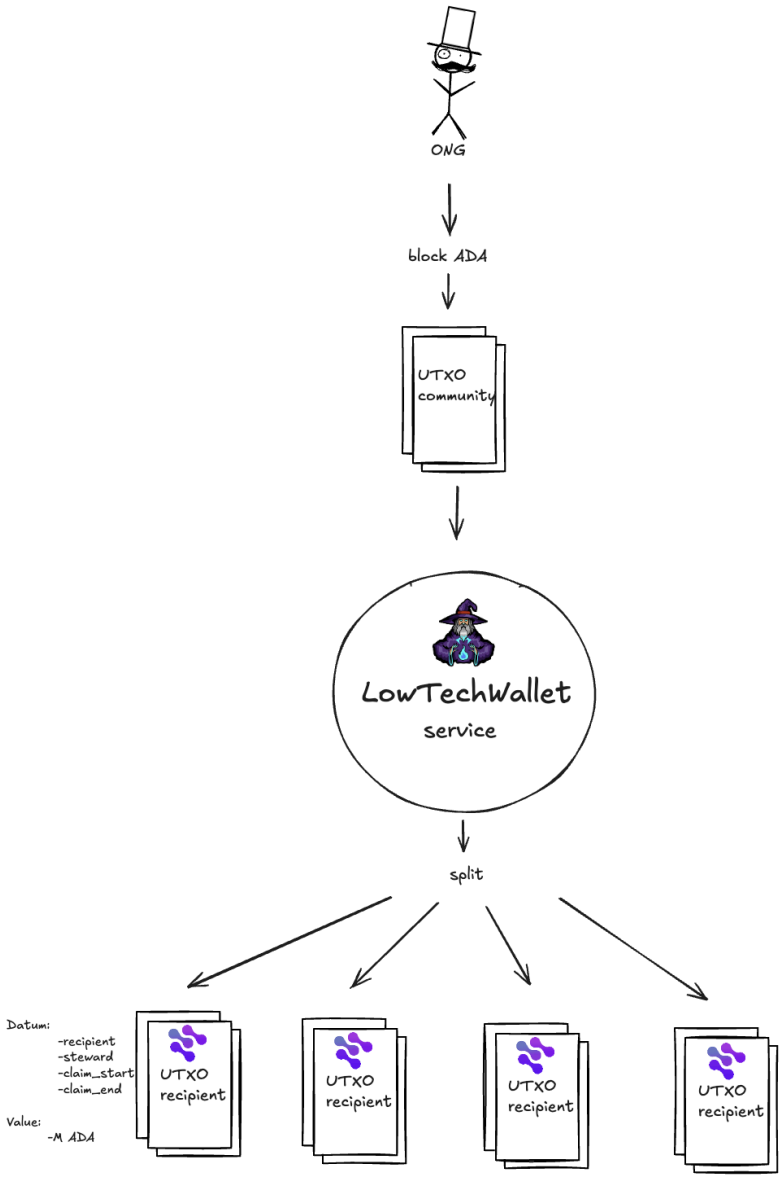
es	**ders**			**NGO** = *Steward* **Beneficiary** = *Recipient*
0. Local SMS	Africa Talking <u>KotaniPay</u> (Existing Partner)	Direct relationship with service provider. Existing partnership in Ghana makes extension to Malawi straightforward.	• Local SMS messaging • Local SMS plus alternative off-ramp	*LTW / SMS Service*
1. NGO Partner Portal	NGOs/ Humanitarian Agencies: Aid distributors	Transparent, efficient aid delivery with automated reporting, enabling NGOs to extend donor contributions further and demonstrate measurable impact. Smart contract automation handles secure fund distribution while maintaining an intuitive interface that requires no blockchain expertise.	Managed Services for Partners: • Customization: Localization features and smart contract-based capabilities for custom requirements • Onboarding: Streamlined onboarding and efficient cash disbursement • Dashboard: Tools for funds traceability and management • Clawback and/ or Claimable Balances Functionality: Asset recovery for enhanced control	**1. Creation of Community Vouchers** • **Context**: NGO allocates funds to a community. • ADA is locked into a **community UTxO** managed by the LTW service (fund custodian). • This UTxO requires only the LTW service control key (no smart contract). • LTW Service subdivides the UTxO into individual voucher UTxOs, each locked with a **Validator script**. **1.1 Late Claim (via LTW Service)** • **Context**: Voucher not consumed before

				expiry. ● Steward initiates transaction. ● Validator checks for steward's signature and authorizes reclaim outside the validity window.
1.1 Program Outreach & ID Verification	NGO/ Aid program sponsor Malawi Government	Greater trust and alignment with NGOs through end-to-end traceability, unlocking additional funding and improving program oversight. Prevent overlap of programs	● Outreach ● Identity verification for program oversight and duplicate prevention	*LTW Service / NGO Dashboard*
2. Beneficiary Low Tech Wallet	Local Community | Smallholder Farmers	Direct, equitable access to aid through simple SMS wallets, reducing barriers to participation and improving financial inclusion.	SMS-enabled offline wallet: ● Account Abstraction: Beneficiary account setup via SMS and in bulk. ● Payment System: Internet-free funds transfer; no SMS credit needed. ● Sponsored Accounts: Seamless asset reception with NGO-funded account setup.	**2. Creation of Individual Vouchers** ● **Context**: Recipient receives a dedicated voucher. ● Each voucher = **UTxO with a spend-type Validator**. ● **Datum fields**: ○ Recipient (authorized spender address) ○ Steward (recovery

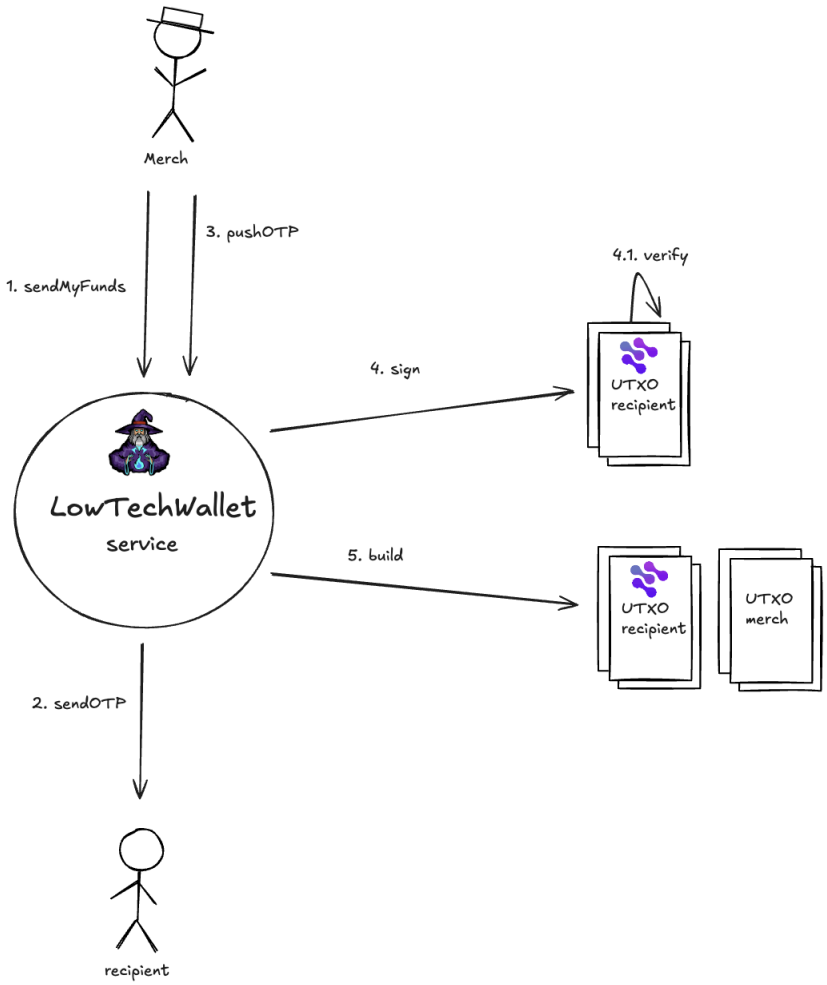
			• Account Management: Simple SMS-based fund and account balance checking • Customization: Tailored smart contract capabilities based on NGO requirements	address) ○ Validity start time ○ Expiration time • **Validator logic**: ○ Validates that spending occurs by the recipient within the time window. ○ If expired, allows the steward to reclaim funds.
3. Merchant Payment	Local Shops: Merchants accepting vouchers	Faster, reliable payouts via vouchers with reduced cash-handling risk, or hefty fees (~6%) from mobile money creating new opportunities to grow their business.	Merchant Services Capabilities: • Reverse Payment Functionality: The merchant initiates the payment process, and the user confirms the transaction amount, simplifying blockchain transactions for users with no knowledge of digital asset transactions • Secure Transactions: Reliable, real-time	**3. Voucher Redemption by Recipient** • **Context**: Recipient spends ADA at a merchant. 1. Merchant initiates payment request on portal (e.g., *30 ADA for goods*). 2. Backend generates OTP and sends via SMS to recipient. 3. Recipient provides OTP to merchant. 4. Merchant enters OTP in portal.

			account balances and low-cost transfers	5. Backend validates OTP and signs transaction via custodial wallet. 6. Validator enforces spending rules, then executes: ○ Recipient's voucher UTxO consumed. ○ New UTxO → Merchant (purchase value). ○ New UTxO → Recipient (remaining balance, if any). **3.1 Partial Expenditure** ● **Context**: Recipient spends less than total voucher value. ● Voucher UTxO split into: ○ Merchant UTxO (spent amount). ○ Recipient UTxO (remaining balance).

1. Creation of Community Vouchers and Creation of Individual Vouchers



2. Voucher Redemption by Recipient



3.3 Tech Stack and Tools

The solution leverages a combination of blockchain technologies, backend services, databases, and communication tools:

Cardano

- Cardano proof-of-stake blockchain platform
- Aiken Smart Contracts
- Vodka: Aiken Development Library
    - Cocktail: On-chain validation utilities
    - Mocktail: Unit test utilities
- Cardano Testnet Faucet
- CardanoScan: Blockchain Explorer
- Laceanatomy: Transaction visualization
- cardano-cli
- cardano-node

FrontEnd

- Ionic
- Angular
- TypeScript
- Mesh SDK (@meshsdk/core, @meshsdk/core-csl)

BackEnd

- Express.js (Backend Framework)
- Axios (HTTP client for backend communication)
- PostgreSQL (Relational Database)
- FerretDB (MongoDB-compatible layer running over PostgreSQL)
- Docker (Containerization)
- SMS Provider
- Low Tech Wallet (LTW) Service

License

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https://gitlab.com/xcapit-foss/app/-/blob/e5c8680dac6db148e6b3f8094c896d2cf864e7b5/LICENSE

Network & Technology Partners

Network/ Partner	Advantage	Capabilities
Cardano Blockchain & Aiken Smart Contract Platform	Leading L1 Blockchain Decentralized Network	• Traceability • Deterministic fees based on transaction size • Deferred Payments
Cardano Foundation/ **Emurgo Labs**	Deep expertise, years of hands-on experience, and a track record of driving blockchain innovation.	• Specialists in Cardano blockchain with direct support.
Xcapit Solution Makers	Proven smart contract development expertise. Prior experience building and testing a low-tech wallet in	• Blockchain Application Development • Smart Contract Engineering • Open-Source Solutions • Low-Tech Wallet

	remote settings. Building for emerging markets since 2017.	Infrastructure
KotaniPay	Local provider with deep regional expertise in digital assets.	● Local SMS Provider ● Alternative off-ramp

3.4 Blockchain Functionality

The solution leverages the Cardano EUTxO model and Aiken Smart Contracts to ensure non-custodial, traceable delivery of aid.

1. Smart Contracts: We deploy a custom Validator Script that enforces strict spending rules:
    a. Identity Verification: The Datum structure stores the beneficiary: VerificationKeyHash, ensuring only the intended recipient can unlock the funds via their SMS signature.
    b. Time-Lock Logic: Parameters claim_start and claim_end (POSIX Timestamps) automatically enforce validity periods, preventing premature use or enabling fund recovery after expiration.
    c. Role Management: The Redeemer structure distinguishes between BeneficiaryAction (spending) and StewardAction (recovery/admin), maintaining a clear separation of duties.
2. UTxO Model Usage (Traceability): Each aid package acts as an individual, traceable UTxO containing the specific amount of ADA and the logic described above. This allows for granular auditing of every single aid packet rather than a pooled account balance.
3. Security & On-Chain Verification:
    a. Signature Verification: The contract utilizes cryptographic checks (checking extra_signatories) to validate that the transaction is signed by the correct private key (managed via the Low Tech Wallet infrastructure) without the user needing internet access.
    b. Double-Spend Protection: We rely on Cardano's native EUTxO guarantees; once an AID UTxO is consumed at a merchant, it ceases to exist, making fraud mathematically impossible.

4. Weekly Workplan

Features/ Timeline	**Week 1-2** Define the challenge, confirm the scope, and finalize your development approach.		**Week 3-6** Move into prototyping sprints.		**Weeks 7-10** Gather feedback from stakeholders, iterate, and refine the solution.		**Weeks 11-12** Finalize a full MVP and prepare a pilot-ready version.	
Features	**Focus / Category**	**Baseline Metric**	**Focus / Category**	**Baseline Metric**	**Focus / Category**	**Baseline Metric**	**Focus / Category**	**Baseline Metric**
0. Local SMS • Local SMS messaging • Alternative off-ramp	Apply for SMS shortcode OR extend KotaniPay partnership to Malawi.	Send application for local shortcode OR Extend our KotaniPay to Malawi	Follow-up on application for local shortcode OR Extend integration with KotaniPay	Follow-up on application for shortcode OR Extend our KotaniPay to Malawi	Test local SMS on low tech phones with 5-10 users (UNDP team on the ground)	Local SMS test results summary	SMS optimization & readiness	Local SMS function implemented
1. NGO Partner Portal • Program rules • Funding • Sponsored accounts • Claimable balances • Account	Define inputs and process for NGO disbursements.	**Testnet: Creation of Community Vouchers (UTxO)** Font-end: Define	Build out solution for program requirements, develop smart contracts for funding, and disbursement modules	**Testnet: UTxO expiration/ claim:** • **UTxO burn** • **Creation of**	Onboard NGO & run account creation and disbursement test. Build out operational dashboar	Font-end / back-end: Dashboard	Streamline system and deliver operational dashboard	Mainnet Launch Mainnet Test & Pilot Readiness

management • Dashboard		Dashboard *Capture*: UTxO NGO -> UTxO Community		**Individual Voucher + Smart Contract** *Capture*: Exp UTxO -> New UTxO + SmartContract	d			
1.1. Program Verification (Malawi Gov/ NGO) • Program oversight • ID verification	Outreach and stakeholder alignment. Showcase system architecture & advantages	tbd	Define role for pilot and map requirements for program/system integration	tbd	Light integration for pilot	tbd	Test optimization and readiness	
2. Beneficiary Low Tech Wallet (Smallholder Farmers) SMS-enabled offline wallet: • Account setup via	Define ID requirements, local SMS integration, and wallet creation.	Define Smart Contract in Aiken *Capture*: Doc define Validator	Build smart contract/ wallet solution	**Testnet:** • **UTxO Split** • **Creation of**	Test onboarding & redemption flows	Testnet: Post transaction UTxO creation (balance) *Capture*: UTxO Recipient + Smart	Outreach and setup at starter location: District of Dowa in the Central Region.	

SMS • Voucher and fund transfers • Sponsored accounts • Simple account management (funds availability, accept payment)		Datum Validator Purpose		**Individual Vouchers (UTxO + Smart Contract)** *Capture*: UTxO Community -> UTxO Recipient + Smart Contract		Contract consumed -> New UTxO Recipient + Smart Contract consumed		
3. Merchant Payment (Small Shops) • Merchant transactions • Voucher redemption confirmed via SMS • Voucher payments	List merchant requirements and Identify key local merchant(s). Outreach to key merchant(s).	Define Merchant actions/ OTP rules Capture: Doc OTP success / failure	Define Merchant payments processes	Testnet: UTxO consumption/ UTxO split *Capture*: UTxO Recipient + Smart Contract consumed -> New UTxO merch	Onboard merchant in Dowa.	Testnet: ***Merchant*** UTxO consumption trigger *Capture*: UTxO Recipient + Smart Contract consumed -> New UTxO merch	Test voucher payouts & fund transfers	

5. Final Deliverables

5.1 Solution Demo (PoC)

Please, find attached a video demo of the current working solution:shelter-demo-cardano.mov

5.2 Testing & Feedback Documentation

Summary of Testing Activities: Prior to field deployment, the technical team conducted comprehensive laboratory testing to ensure system security and stability on the **Cardano Preview Testnet**. We executed strict scenario testing based on three core User Stories:

- **Aid Assignment (US-01):** We verified the **"Assign AID"** workflow, confirming that the NGO (Steward) can successfully lock funds into the smart contract attached to a specific beneficiary's hash and a defined validity period.
- **Fund Consumption (US-02):** We validated the **"Consume AID"** logic, ensuring that beneficiaries can only spend the funds if the transaction is signed by their unique key and occurs strictly within the active claim window.
- **Fund Recovery (US-03):** We tested the **"Recover Expired AID"** mechanism, proving that the NGO can reclaim unused funds only after the expiration timestamp (claim_end), preventing capital from remaining permanently locked.

Challenges & Resolutions: During the development phase, detailed in the Debugging and Testing Report V1.0, we addressed critical technical challenges:

- **Time-Lock Precision:** Initial challenges with transaction validity intervals were resolved by implementing explicit safety margins for invalidBefore and invalidHereAfter slots, ensuring beneficiaries can only claim funds during the active window.
- **Error Clarity:** We moved from generic failure states to explicit trace messages. This ensures that if a transaction fails (e.g., due to a missing signature or expired timestamp), the system provides clear feedback (e.g., [Contract Error]: StewardAction, must be signed with steward key) for rapid troubleshooting.

System Feedback: The system successfully enforces security rules. Stress tests confirmed that the smart contract correctly rejects unauthorized actions, such as attempts to spend funds after the expiration date or transactions missing the required cryptographic signatures.

5.3 Feedback Integration Summary

Despite coordination challenges with the UNDP Malawi office, we used our learnings from prior deployments to iterate and strengthen the solution in two concrete areas: beneficiary usability and NGO visibility/impact reporting.

First, we refined the SMS user journey to reduce confusion and increase completion rates. This included simplifying message wording, clarifying call-to-action steps, improving the sequencing of prompts, and adding clearer confirmations and error guidance. The goal of these changes is to lower barriers for beneficiaries using basic phones, so they can successfully access and use the support they receive with minimal assistance.

Second, based on stakeholder feedback, we increased our investment in a more robust NGO-facing impact dashboard. We expanded the reporting layer to provide clearer end-to-end traceability and operational metrics, enabling programme teams to monitor onboarding and disbursement progress, identify bottlenecks, and generate evidence for accountability. This iteration strengthens transparency and trust by making performance and outcomes easier to track, explain, and audit across the full distribution cycle.

5.4 Pilot-Ready Solution

Repository: https://github.com/xcapit/cardano-voucher

Demo video: shelter-demo-cardano.mov

5.5 Confirmed Pilot Partnership(s)

Based on the accelerator timeline and ongoing coordination, the following partnership structure is proposed:

Primary Partnership Track: UNDP Malawi Office

- Status: Active coordination with the UNDP Malawi Country Office as Challenge Owner
- Focus Area: Agricultural Aid Distribution (Affordable Inputs Programme - AIP) and Social Cash Transfer Programme (SCTP)
- Target Region: District of Dowa in the Central Region, selected for pilot accessibility and existing government infrastructure
- Beneficiary Pool: Initial cohort of 200-500 smallholder farmers from AIP beneficiary lists

Alternative Partnership Track: UNDP Georgia Office

- Status: Preliminary discussions initiated as contingency pathway
- Rationale: Georgia offers favorable regulatory environment for blockchain pilots with existing digital government infrastructure
- Potential Use Case: Agricultural subsidy distribution or social protection programs

Technical Partners (Confirmed)

- Cardano Foundation: Technical advisory, ecosystem support, and co-marketing
- KotaniPay: Local SMS gateway integration (existing partnership from Ghana deployment, extension to Malawi in progress)
- Africa's Talking: Backup SMS provider with established Malawi coverage

5.6 Stakeholder Alignment Matrix

Still to be confirmed according to previous point disclosure.

The following matrix defines roles, responsibilities, and coordination mechanisms for all pilot stakeholders:

Stakeholder	Role	Key Responsibilities	Involvement	Engagement	Contact
UNDP Malawi Office	Challenge Owner	Strategic direction, government liaison, beneficiary access, regulatory navigation, impact reporting	Lead	MOU	Khama
Xcapit	Solution Maker	Platform development, smart contract deployment, SMS integration, technical support, iteration cycles	Lead	Contract	José Trajtenberg (CEO) Joel Dellamaggiore (Blockchain Dev)
Cardano Foundation	Network Partner	Technical advisory, mainnet deployment support, ecosystem connections, co-funding exploration	Support	Partnership	TBD
Ministry of Agriculture (Malawi)	Government Sponsor	AIP beneficiary lists, policy alignment, district coordination, official endorsement	Support	Gov Letter	Via UNDP
Ministry of Gender & Social Welfare	Government Sponsor	SCTP beneficiary data, social protection alignment, vulnerability targeting criteria	Support	Gov Letter	Via UNDP
KotaniPay / Africa's Talking	Tech Partner	SMS gateway, local shortcode, USSD fallback, off-ramp integration	Support	Service Contract	Existing
Local Merchants (Dowa District)	Service Provider	Voucher redemption, goods/services provision, transaction confirmation	Participant	Onboarding	Via NGO

Coordination Mechanisms

- Weekly sync calls between Xcapit and UNDP focal point during active development phases
- Bi-weekly steering committee updates with all Lead stakeholders
- Dedicated Slack/Teams channel for real-time technical coordination
- Monthly written progress reports aligned with accelerator milestones

Shared dashboard access for pilot monitoring (read-only for stakeholders)

5.7 Budget Proposal (Pilot)

The following budget breakdown, according to the previous points confirmation, covers the pilot implementation phase (3-6 months post-accelerator):

Budget Line Item	Unit / Description	Qty	Unit Cost	Total (USD)	Funding
Technical Lead (Blockchain)	Monthly retainer	4	$4,500	$18,000	Pending
Backend Developer	Monthly retainer	4	$3,500	$14,000	Pending
Project Manager	Monthly (part-time)	4	$2,000	$8,000	Pending
SMS Gateway (KotaniPay)	Monthly + per-message	4	$800	$3,200	Pending
Cloud Infrastructure (AWS)	Monthly hosting	4	$400	$1,600	In-kind
Cardano Mainnet Fees	Transaction fees (est.)	1	$500	$500	Pending
Local Field Coordinator	Monthly (Malawi-based)	4	$1,200	$4,800	Pending
Beneficiary Onboarding	Per beneficiary (200)	200	$5	$1,000	Pending
Merchant Onboarding	Per merchant (10)	10	$50	$500	Pending
Travel & Field Visits	Trips to Dowa	3	$1,500	$4,500	Pending
User Testing Sessions	Facilitated sessions	4	$300	$1,200	Pending
Security Audit (Smart Contract)	Third-party audit	1	$5,000	$5,000	Cardano F.
Contingency (10%)	Unforeseen expenses	1	$6,230	$6,230	Pending
TOTAL ESTIMATED COST				**$68,530**	

Funding Sources Strategy

- UNDP Accelerator Grant: Target $25,000-35,000 for core pilot operations
- Cardano Foundation: In-kind technical support + potential $10,000-15,000 for security audit and mainnet fees
- Xcapit In-Kind Contribution: Estimated $20,000 value (reusable platform components, accumulated R&D, operational playbooks from Peru/Kenya)
- Co-financing from NGO Partner: Field operations support, beneficiary mobilization

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

- SDG 09: Industry, Innovation and Infrastructure
    - Target: Significantly increase access to information and communications technology and strive to provide universal and affordable access to the Internet in least developed countries by 2030.
    - Indicator: Number of individuals reached who utilize Basic Feature Phones (non-smartphones) to access financial aid services via SMS infrastructure.
- *SDG 10: Reduce Inequalities*
    - *Target: By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion, economic or other status.*
    - *Indicator: Proportion of people living below 50 percent of median income, by age, sex, and persons with disabilities.*

5.8.2 Key Performance Indicators (KPIs)

KPI	***Measurement Method***	***Target / Threshold***	***Notes***
Number of successful validator transactions	Blockfrost API / Cardano Explorer logs	95%+ success rate	
Active users onboarded	LTW Service	200 users	Tracked via wallet addresses
Number of people reached who do not have access to formal financial products or services	LTW Service	200 users	Directly measures financial inclusion enabled by the SMS Wallet infrastructure.
Proportion of people living	Demographic breakdown of the	100% of target group	Ensures funds are strictly routed to the identified

below 50% of median income, by age, sex, and persons with disabilities	"Beneficiary Upload" CSV files provided by the NGO.		vulnerable populations verified by the NGO.

5.8.3 Monitoring & Reporting Plan

The monitoring strategy employs a Dual-Layer Verification approach. We cross-reference off-chain user data from the LTW Service (Backend) with immutable on-chain records from the Cardano Blockchain to ensure 100% data integrity and transparency.

Frequency: Quarterly Impact Reports

- Responsible Party: Project Manager
- Format:
    - Technical: On-chain execution logs (Blockfrost) compared against backend activity logs.
    - Impact: PDF/CSV Summaries generated via direct on-chain queries visualizing funds disbursed vs. claimed.
    - Off-chain: information provided by the NGO working on the field implementation with end-beneficiaries.
- Tools:
    - LTW Backend Service: Primary source for user activity logs, SMS history, and beneficiary metadata.
    - Blockfrost API: Used to query the Cardano Ledger for independent verification of transaction finality and UTxO status.
    - LTW Database: Primary Administrative Interface. Currently used via direct query/CLI to monitor user onboarding, system status, and generate data exports for the NGO.
    - NGO´s Surveys: end-beneficiaries information provided to the NGOs working on the field.

5.8.4 Risks Assessment & Mitigation Plan

Risks	Mitigation
Partial or Delayed information from the entity disbursing funds.	Impact dashboard with the information to do bulk creation and transfer
Multi-party organizations taking part of the disbursement and fractional information.	Implementation of the Rahat Dashboard standards to aggregate information from multiple participants into a single,

	interoperable source of truth compatible with global humanitarian metrics.
Incomplete surveys from the people in the field.	Listing a form that needs to be completed to be eligible for the support or aid.

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
• **Pilot launch (Malawi or Georgia):** beneficiary registry, bulk wallet creation, bulk disbursement, SMS flows, and monitoring dashboard. • **Shelter V2 rollout:** deploy the improved UX and operational tooling created from prior pilot feedback. • **Evidence baseline:** define KPIs/targets with UNDP, implement data collection, and run pre- and post-pilot feedback cycles.	• **Operational scale:** increase beneficiary volume and transaction frequency; expand merchant coverage and settlement/offramp reliability. • **Product hardening:** resilience, security reviews, localization, admin workflow improvements, and automation for recurring disbursements. • **Replication package:** “country office starter kit” (technical deployment, training materials, governance, and reporting templates). • **Ecosystem collaboration:** build joint modules with Cardano Solution	• **Multi-country programme**: replicate in multiple UNDP offices, supported by a shared framework for traceability and impact reporting. • **Sustainability model:** long-term maintenance, partner enablement, and co-financing strategy with country offices and donors. • **Ecosystem growth**: open, modular components and integrations to accelerate adoption within and beyond UNDP programmes.

	Makers (identity, attestation, analytics, compliance tooling).	

5.9.2 Budget for Phased Implementation (post-acceleration)

Below is a phased, high-level budget structure aligned with the roadmap. Amounts will be finalized with UNDP once the pilot country, partner readiness, beneficiary volume, and compliance requirements are confirmed. The budget is organized to separate core platform work (largely reusable across countries) from country deployment costs (which vary by context).

Short Term (0–6 months): Pilot launch + Shelter V2 rollout + evidence baseline

Budget range: USD 50,000 – 100,000

Cost categories

Pilot implementation and technical delivery (40–55%)

Beneficiary registry setup, bulk wallet creation, bulk disbursement configuration
SMS flow configuration and localization for the pilot context
Deployment, monitoring, incident response readiness
Product updates (Shelter V2 rollout) and UX iteration (15–25%)
Deploy improved flows, copy optimization, error handling, and usability fixes

Admin workflow refinements based on field constraints

Impact dashboard and reporting (10–20%)
Monitoring dashboard setup, traceability views, KPI tracking, reporting templates
Testing, M&E baseline, and feedback cycles (10–15%)
Pre- and post-pilot surveys, interviews, structured observations, pilot retrospective
Security, QA, and compliance readiness (5–10%)
Code review, basic security hardening, operational controls, documentation
Country operations and coordination (5–10%)
Field coordination with implementing partners, training sessions, materials

Expected outcome of this phase

Pilot successfully launched (Malawi or Georgia), Shelter V2 deployed, baseline KPIs defined and measured, and a validated backlog for the next iteration.

Medium Term (6–18 months): Operational scale + hardening + replication package + ecosystem modules

Budget range: USD 150,000 – 400,000

Cost categories

Scaling operations and partner expansion (25–40%)
Increase beneficiary volume, transaction frequency, merchant coverage
Improve settlement/offramp reliability and operational playbooks
Product hardening and reliability (20–30%)
Resilience upgrades, security reviews, performance improvements
Automation for recurring disbursements and operational tooling
Replication toolkit ("country office starter kit") (15–25%)
Deployment scripts, governance templates, training curriculum, documentation
Standardized KPI and reporting framework for new country offices
Interoperability and ecosystem collaboration (10–20%)
Joint modules with Cardano Solution Makers (identity/attestation/analytics/compliance)
Integration and testing cycles for partner components
Ongoing user research and iteration (5–10%)
Periodic testing rounds, issue triage, UX improvements tied to adoption metrics
Expected outcome of this phase

A hardened platform ready for repeated deployments, with a replication package enabling faster onboarding of additional UNDP offices and ecosystem integrations.

Long Term (18+ months): Multi-country programme + sustainability model + ecosystem growth

Budget range: USD 500,000 – 800,000+ (depending on number of countries and beneficiary scale)USD 500,000 – 800,000+ (depending on number of countries and beneficiary scale)

Cost categories

Multi-country rollout and programme support (35–55%)
Deployments across multiple UNDP offices, regional coordination, training, support
Long-term maintenance and governance (15–25%)
SLA-style support, monitoring, upgrades, incident management, partner enablement
Ecosystem expansion and modularization (10–20%)
Open, reusable modules; standardized APIs; integration pathways for third parties
Evaluation, learning, and donor reporting (10–20%)
Cross-country impact reporting, audits, evidence packaging for co-financing/donors

Expected outcome of this phase

A sustainable, repeatable multi-country deployment model with long-term operational capacity, co-financing pathways, and broader Cardano ecosystem participation.

Notes and assumptions:

Final budget depends on: pilot country selection, beneficiary count, number of merchants, required integrations (KYC/offramp), localization needs, and compliance constraints.

The budget primarily covers platform delivery, deployment, testing, monitoring, and reporting; direct transfer value to beneficiaries is typically handled by programme funds and is not included unless explicitly required.

Co-financing/in-kind may include existing Shelter V2 capabilities, reusable components, and prior deployment learnings that reduce time-to-delivery and cost per additional country rollout.

Short Term (0-6 mo)	**Pilot launch + Shelter V2 rollout + evidence baseline**	6 months	**USD 50,000 – 100,000**
Medium Term (6-18 mo)	**Operational scale + hardening + replication package + ecosystem modules**	12 months	**USD 150,000 – 400,000**
Long Term (18+ mo)	**Multi-country programme + sustainability model + ecosystem growth**	Ongoing	**USD 500,000 – 800,000+ (depending on number of countries and beneficiary scale)**

5.9.3 Financial Sustainability Model

Post-pilot, the solution will operate under a **B2B / programme-based model** tailored to UN agencies, NGOs, and implementing partners. Income generation is designed to be predictable for programme budgets and scalable across countries.

Pricing components

- **Implementation / onboarding fee (one-time):** covers configuration for the country context, integration work (e.g., offramp/merchant partners), localization, training, and go-live support.

- **Platform subscription (monthly / annual):** access to the admin console, beneficiary registry, bulk wallet creation, bulk disbursement tooling, monitoring dashboard, and reporting templates.

- **Transaction-based fee (variable):** a small fee per beneficiary created and/or per disbursement/transaction processed, aligned with volume and operational load.

- **Optional modules (add-ons):** enhanced analytics and proof-of-impact reporting, identity/attestation modules, compliance tooling, or advanced integrations requested by the programme.

When it kicks in

- During the pilot, fees can be reduced or partially covered by programme funding and/or grants to de-risk adoption.

- **Commercial pricing starts immediately after pilot completion** (or at the point of scaling to the next cohort / district), once the solution is validated and the implementing partner moves into an operational rollout. At that point, the subscription + per-transaction model becomes the default, with multi-country discounts for UNDP regional programmes.

5.9.4 Partnership and Co-Financing Commitments

The project will be supported through a combination of technical partners, institutional partners, and in-kind contributions, structured to reduce deployment risk and accelerate replication.

Institutional / programme partners

- UNDP country office (Malawi or Georgia) + implementing partners: programme alignment, beneficiary selection and safeguarding processes, field coordination, user research access, and operational validation.

Technical / ecosystem partners

- Cardano Solution Makers: co-development of ecosystem modules (identity, attestations, analytics, compliance tooling), security best practices, and technical support for Cardano-native deployments.

- Payments / offramp / settlement partners (country-specific): enable cash-out or merchant settlement flows, support operational reconciliation, and ensure funds can be used locally in a compliant manner. Eg. Kotani Pay; Seevcash, etc.

Our contribution (in-kind / co-financing)

- Reuse of existing infrastructure (SMS orchestration, wallet lifecycle, bulk disbursement, monitoring dashboard) and deployment playbooks from prior implementations.

- Engineering, product, and UX capacity to implement Shelter V2 and iterate during pilot cycles, reducing total cost and time-to-deploy.

5.9.5 Fundraising Plan

We will pursue a blended financing strategy combining ecosystem funding, donor grants, and programme budgets.

Target sources

- Cardano ecosystem funds: Catalyst and ecosystem-aligned funds for technical development, security reviews, and open modules (identity/attestation/analytics).

- UN / donor grants: innovation and digital public goods funding streams aligned with financial inclusion, humanitarian response, social protection, and public sector digitization.

- Programme financing: once the pilot demonstrates operational value, UNDP country offices and implementing partners can fund scaling via subscription + per-transaction fees as part of programme budgets.

- Strategic partners / investors: mission-aligned impact investors or corporate partners interested in fintech inclusion and proof-of-impact infrastructure.

Timeline

- 0–3 months: pilot scoping and grant applications aligned to the pilot country; ecosystem funding proposals for Cardano module work.

- 3–6 months: secure funds for scaling to the next cohort and for security/performance hardening; prepare replication package. Begin institutional donor outreach.

- 6–18 months: multi-country fundraising for replication, potentially via regional programmes and co-financing with donors and country offices. Close seed round or secure anchor institutional grant

5.10 Legal & Compliance Considerations

Key considerations will be addressed with a country-specific compliance approach, coordinated with UNDP and implementing partners:

- **Data protection and privacy:** beneficiary data minimization (collect only what is needed), secure storage, access controls, retention policies, and alignment with applicable laws (e.g., local data protection frameworks and GDPR principles where relevant).

- **Safeguarding and responsible use:** processes to protect vulnerable beneficiaries, consent and communication clarity, grievance mechanisms through implementing partners, and safe operational support.

- **KYC/AML and financial regulation (context-dependent):** where required, reliance on regulated partners (telcos, mobile money, banks, or licensed PSPs) for onboarding/offramp; documentation of compliance responsibilities by role.

- **Consumer protection and dispute handling:** clear messaging, confirmation steps, error handling, and defined processes for reversals/issue resolution where the programme design requires it.

- **Smart contract and security assurance:** code review, testing, and release controls; audit strategy proportionate to deployment risk; incident response procedures.

- **Cross-border funding and sanctions screening (if applicable):** ensure that disbursement sources and settlement flows meet UNDP policies and local banking requirements.

Malawi Regulatory Environment

- Reserve Bank of Malawi (RBM): Primary financial regulator; no specific cryptocurrency framework as of 2024, but digital payment services require licensing
- Malawi Communications Regulatory Authority (MACRA): Oversight for SMS-based services and shortcode registration
- Data Protection: Malawi enacted the Data Protection Act in 2024; compliance required for beneficiary data handling
- AML/CFT: Anti-money laundering requirements apply to any value transfer system

Compliance Strategy

- Voucher Classification: Structure vouchers as prepaid instruments/digital tokens with restricted merchant acceptance rather than general-purpose currency
- Custodial Model: Operate under NGO/government partner's existing financial licenses where possible
- KYC/KYB Requirements: Beneficiary verification through existing government ID systems (National ID) and NGO enrollment processes

5.11 Visibility Assets / Communication Materials

Slide 1: "Aid & Benefits Distribution Without Connectivity Barriers"

- *Problem: beneficiaries often lack smartphones, internet, or bank accounts; programme teams need traceability and accountability.*
- *Solution: SMS-based flows + bulk wallet creation + bulk disbursement + real-time monitoring dashboard.*
- *Outcome: reduced operational friction, improved inclusion, and stronger transparency from allocation to final usage.*

Slide 2: "Trust, Traceability, and Proof of Impact"

- *End-to-end transparency: beneficiary registry → disbursement → usage/settlement reporting.*
- *Evidence layer: dashboard KPIs, audits, and programme reporting templates.*
- *Scalable model: replication toolkit for additional UNDP offices; modular integrations with Cardano Solution Makers.*

Short storyline (for a one-minute pitch)
"UNDP programmes often operate where connectivity and formal financial access are limited, yet the need for accountability is highest. Our solution enables programme teams to onboard beneficiaries using only phone numbers, create wallets at scale, and execute bulk disbursements while beneficiaries interact through simple SMS flows. On the back office side, UNDP and partners gain a monitoring and impact dashboard that provides traceability and reporting from end to end. After validating the approach through pilots and production deployments, we are now making the solution Cardano-ready and packaging it for replication across multiple UNDP country offices."

5.12 Pitch Deck

Deck(Google Drive): Shelter-on-Cardano-Deck.pdf

Deck(Gamma): Shelter-on-Cardano-Deck

Shelter in action(Video - Google Drive): shelter-on-cardano.mp4

5.13 Final Demo Presentation

Demo Final Presentation(Video - Loom): Introducing Shelter on Cardano: A Solution for Aid Transparency in Malawi

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	**Week 1**
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	**Week 2**
3. Solution Overview (core features, tech stack, blockchain functionality)	**Week 2**
5.1 Solution Demo (PoC)	**Week 3**
5.2 Testing & Feedback Documentation	**Week 5**
5.3 Feedback Integration Summary	**Week 6**
5.4 Pilot-Ready Solution	**Week 6**
5.5 Confirmed Pilot Partnership(s)	**Week 7**
5.6 Stakeholder Alignment Matrix	**Week 4**
5.7 Budget Proposal (Pilot)	**Week 5**
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	**Week 5**
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	**Week 7**
5.11 Legal & Compliance Mapping	**Week 4**

5.12 Visibility Assets / Communication Materials	**Week 6**
5.13 Pitch Deck	**Week 8**